

Training recipes and graph-native learning

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Developer guide

Training is separate from the graph

`TrainSpec` describes standard learning choices without embedding a training loop in graph structure. It refers to named outputs and parameters in one network.

```
from tools.snnlang import training

recipe = training.TrainSpec(
    objectives=[training.CrossEntropy(
        prediction=readout,
        target="digit",
    )],
    parameter_groups=[training.ParameterGroup(
        name="feedforward",
        parameters=readout.parameters,
        lr=1e-3,
    )],
    optimizer=training.AdamW(weight_decay=1e-4),
    epochs=20,
)

bundle = snn.compile(net, training=recipe)
```

The recipe vocabulary includes objectives, named parameter groups, learning rates, frozen parameters, optimizer settings, regularizers, stop-gradient boundaries, epoch count, gradient clipping, and surrogate configuration. Every graph parameter must belong to exactly one group. Trainable groups require a positive finite learning rate; frozen groups require zero. Compilation records canonical resolved trainable and frozen sets plus a per-parameter learning-rate map so backends never infer ownership from naming conventions.

Current boundary

Training recipes compile, and a narrow one-layer MNIST PING recipe can be translated into the legacy trainer. The typed graph API executes deterministic minibatch AdamW trajectories for cross-entropy and spike-budget objectives through named trainable/frozen groups. Named input and target files retain their digests in the execution protocol; seed-derived epoch permutations and checkpointed batch position support exact mid-epoch CPU resume. Requests never fall back silently to legacy execution.

Deterministic initialization, differentiable-reachability checks, recurrent trainability, variable-rate protocols, spike-budget regularization, selected and final checkpoints, optimizer-state replay, and exact CPU resume are implemented. Accelerator stochastic-state support and layered legacy-versus-graph conformance tests remain.

API reference

TrainSpec

```
training.TrainSpec(  
    objectives,  
    parameter_groups,  
    optimizer,  
    regularizers=(),  
    stop_gradients=(),  
    epochs=1,  
    gradient_clip=None,  
    surrogate=training.FastSigmoid(slope=1.0),  
    presentation_duration=200 * snn.ms,  
)
```

The specification is data only. `compile` verifies referenced outputs, parameters, regularizer signals, stop-gradient signals, and differentiable routes before writing `training.json`.

The collection's supported surrogate is `training.FastSigmoid(slope=1.0)`. Its slope must be positive and finite. Voltage-gradient dampening is declared per spiking population as `voltage_grad_dampen`; it is also resolved into `training.json` so training provenance contains the complete backward contract. Dampening factors must be positive and finite. The narrow legacy adapter accepts fast-sigmoid slopes and a single shared dampening factor, and rejects richer unsupported combinations explicitly.

Presentation duration is a positive millisecond quantity independent of graph timestep, but it must resolve to an integer number of steps. `SpikeBudgetPenalty` implements the exp022 one-sided quadratic firing-rate ceiling. It converts each population's spike count to a per-presentation population-mean rate in Hz, applies the squared hinge above the ceiling, then averages presentations and layers. This makes the term invariant to batch size, population width, hidden-layer count, and equivalent changes in presentation duration.

Compilation proves a gradient route for every objective and regularizer by walking backward through operations and enabled projections. It intersects the reachable named parameters with the resolved trainable set and respects `StopGradient` boundaries. Frozen-only paths, disabled trainable projections, and barriers with no downstream trainable parameter fail with

the exact reachable and trainable sets. Recurrent parameters are treated structurally, so frozen, trainable initialized, trainable zero-initialized, and trainable small-initialized loop variants use the same vocabulary without name-based exceptions.

Objectives and parameter groups

```
training.CrossEntropy(*, prediction, target, weight=1.0) -> Objective
training.ParameterGroup(parameters, name, lr, frozen=False)
```

`prediction` accepts a signal-like object or identifier. It must resolve to a named graph output. `target` is the external target identifier. Group identifiers are non-empty and unique; groups are non-empty; and every parameter appears exactly once across the groups. Frozen groups use `lr=0`, while trainable groups use a positive finite rate.

Optimizer and regularization

```
training.AdamW(**config) -> Optimizer
training.UpperRatePenalty(*, signal, threshold, strength) -> Regularizer
training.SpikeBudgetPenalty(*, signals, ceiling_hz, strength) -> Regularizer
training.StopGradient.at(signal) -> StopGradient
training.FastSigmoid(*, slope=1.0) -> Spec
```

Optimizer configuration is serialized without running an optimizer. `SpikeBudgetPenalty` is the canonical multi-layer form; `UpperRatePenalty` is its single-signal compatibility spelling. `StopGradient.at` records a graph boundary by signal identifier.

Execution support

`snn.compile(net, training=recipe)` validates and serializes the recipe. `tools.snn.execution.train(ExecutionSpec(executor="graph", training=..., inputs=..., targets=...))` performs one update by default; `options={"updates": n}` repeats the same resolved batch for focused trajectory checks. Setting `options={"epochs": n, "batch_size": b, "shuffle": true}` iterates the sample axis with a deterministic permutation per epoch. `TargetArrayBinding` and `load_target_array_bindings` provide named NPY/NPZ targets with source digests. A training bundle authenticates and supplies its own recipe. The result exposes named gradients, optimizer state, selected/final checkpoints, and exact next-batch state without tensor-position mapping.

Next: Runtime state, checkpoints, and provenance